Crop Yield Prediction Using Machine Learning

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Abstract

This project developed and evaluated machine-learning models for predicting crop yield using rainfall, temperature, fertilizer application, nitrogen, phosphorus, potassium and crop type. The original dataset contained 1,000 observations, and one duplicate was removed to obtain a cleaned dataset of 999 observations.

Exploratory analysis showed that the available predictors had generally weak relationships with crop yield. Five additional variables were engineered to represent nutrient composition, nonlinear temperature effects and the interaction between rainfall and fertilizer.

Multiple Linear Regression and Random Forest Regression were evaluated using both the original and engineered predictor sets, while Extreme Gradient Boosting was evaluated using the engineered predictor set. This produced five modelling experiments. Performance was assessed using Mean Absolute Error, Root Mean Squared Error, the coefficient of determination and five-fold cross-validation.

The engineered XGBoost model achieved the lowest hold-out test RMSE of 2.9219 quintals per acre and the highest test R^2 of 0.0091. However, the engineered Random Forest achieved the lowest test MAE of 2.525 quintals per acre and a slightly better mean cross-validation RMSE of 2.762 ± 0.125. Because the differences were negligible and the Random Forest pipeline had already been successfully integrated into the Streamlit application, it was retained as the deployment model.

The very low test R^2 values across all models showed that the available variables explained little of the variation in unseen crop yields. Model interpretation suggested that rainfall and temperature were relatively more useful than the other predictors, although their absolute predictive contributions remained small.

The deployed application is therefore presented as a technical prototype rather than a validated agricultural decision-support system. The main recommendation is to collect richer agricultural data containing geographic, temporal, soil, irrigation, crop-variety and farm-management information. The findings indicate that improving data quality and relevance is more important than increasing model complexity.

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1 Introduction

1.1 Background

Crop yield is influenced by environmental, biological and farm-management factors, including rainfall, temperature, fertiliser application, soil nutrients, crop variety and crop type. Reliable yield prediction can support agricultural planning, resource allocation, production forecasting and evidence-based decision-making.

This project applies statistical and machine-learning methods to predict crop yield using rainfall, temperature, fertiliser, nitrogen, phosphorus, potassium and crop type.

Multiple Linear Regression was used as the baseline model. Random Forest Regression and Extreme Gradient Boosting were evaluated as nonlinear tree-based methods capable of representing more complex relationships and interactions among the predictors.

Feature engineering was also performed to determine whether derived nutrient, nonlinear and interaction variables could improve model performance.

1.2 Problem Statement

Crop-yield prediction is challenging because agricultural outcomes are influenced by multiple interacting factors. Environmental conditions, nutrient availability, crop characteristics, soil properties and farm-management practices may all contribute to observed yield differences.

The supplied dataset contained only a limited set of these factors. It was therefore necessary to determine how much predictive information could be extracted from the available variables and whether more flexible machine-learning algorithms could improve upon a linear baseline.

The project also required an assessment of which variables were most useful to the selected prediction model. These results were interpreted as model-based associations rather than evidence of causal agricultural effects.

1.3 Project Objectives

The objectives of this project were to:

1. clean, prepare and explore the crop-yield dataset;
2. examine the distributions and relationships among rainfall, temperature, fertiliser, nutrient variables, crop type and crop yield;
3. create engineered variables representing nutrient composition, nonlinear temperature effects and the interaction between rainfall and fertiliser;
4. develop Multiple Linear Regression, Random Forest and XGBoost models for crop-yield prediction;
5. compare original and engineered predictor sets where applicable;
6. evaluate model performance using Mean Absolute Error, Root Mean Squared Error, the coefficient of determination and five-fold cross-validation;

7. interpret the contribution of the available predictors to the selected model while recognising the limitations of model-based impact analysis;
8. provide recommendations for improving crop-yield prediction and agricultural data collection; and
9. deploy the selected model through a Streamlit technical prototype.

1.4 Scope of the Project

The study focused on predicting crop yield using rainfall, temperature, fertiliser, nitrogen, phosphorus, potassium and crop type.

The completed analysis covered:

- data cleaning and preparation;
- exploratory data analysis;
- feature engineering;
- development of linear, bagging and boosting models;
- hyperparameter tuning;
- hold-out and cross-validation evaluation;
- interpretation of the selected model;
- formulation of business and data recommendations; and
- deployment of a Streamlit prediction prototype.

Multiple Linear Regression and Random Forest were evaluated using both the original and engineered predictor sets. XGBoost was evaluated using the engineered predictor set. This produced five modelling experiments.

The findings are limited to the observations and variables contained in the supplied dataset. The dataset did not include geographic, temporal, soil, irrigation, crop-variety or detailed farm-management information.

External weather, soil and satellite data were not integrated because the dataset lacked the geographic and temporal identifiers required for reliable matching.

The deployed application is therefore presented as a technical demonstration of the prediction workflow rather than as a validated agricultural decision-support system.

2 Data and Methodology

2.1 Dataset Description

The original dataset contained 1,000 observations and eight variables. Each observation represented a crop-production record containing environmental conditions, fertilizer application, macronutrient information, crop type and recorded yield.

The variables were:

- **Rainfall (mm):** the amount of rainfall recorded during the farming period, measured in millimetres.
- **Temperature (°C):** the temperature conditions recorded during the farming period.
- **Fertilizer (kg):** the quantity or concentration of fertilizer applied, recorded in kilograms.
- **Nitrogen (N):** the recorded amount of nitrogen present in the fertilizer or soil.
- **Phosphorus (P):** the recorded amount of phosphorus present in the fertilizer or soil.
- **Potassium (K):** the recorded amount of potassium present in the fertilizer or soil.
- **Crop type:** the type of crop produced, including crops such as sorghum, corn and wheat.
- **Yield (Q/acre):** crop yield measured in quintals per acre. This was the target variable.

2.2 Data Cleaning and Preprocessing

A systematic data-cleaning process was carried out to improve data quality and prepare the dataset for exploratory analysis and modelling.

2.2.1 Initial Inspection

The dataset was first inspected to understand its structure, variable types and overall quality. The `info()` and `describe()` functions were used to examine the data types, summary statistics and categorical values.

The raw dataset contained 1,000 observations and eight variables. It included seven numerical variables and one categorical variable, crop type.

2.2.2 Column Name Standardisation

The original column names contained inconsistent spacing, capitalisation and spelling. They were standardised by:

- removing leading and trailing spaces;
- converting all names to lowercase;
- replacing spaces and special characters with underscores; and
- correcting spelling errors such as `temperatue` and `yeild`.

The final column names were:

```
crop_type, rainfall, temperature, fertilizer,
nitrogen, phosphorus, potassium, yield
```

2.2.3 Missing-Value Assessment

Missing values were examined using the `isna()` function. No missing values were found in the dataset, so no imputation or observation removal was required for missing data.

2.2.4 Duplicate Removal

Duplicate observations were identified using the `duplicated()` function. One duplicate row was found and removed to prevent repeated information from influencing the analysis and model-training process.

After removing the duplicate, the dataset contained 999 unique observations.

2.2.5 Data-Type Validation

The numerical variables were converted to appropriate numeric data types using `pd.to_numeric()` with error checking. This ensured that rainfall, temperature, fertilizer, nutrient values and yield could be used correctly during statistical analysis and modelling.

Crop type was retained as a categorical variable.

2.2.6 Categorical-Label Standardisation

Crop-type labels were converted to lowercase and unnecessary spaces were removed. This ensured that different versions of the same crop name were not treated as separate categories.

For example, labels such as `Corn`, `corn` and ` corn ` would all be recorded consistently as `corn`.

2.2.7 Range Validation

The numerical variables were checked for negative or physically implausible values. No negative values were found in rainfall, temperature, fertilizer, nitrogen, phosphorus, potassium or yield.

The observed values were therefore retained for further analysis.

2.2.8 Outlier Assessment

Potential outliers were assessed using box plots and the Interquartile Range method. An observation was considered a potential outlier when it fell below

$$Q_1 - 1.5 \times IQR$$

or above

$$Q_3 + 1.5 \times IQR.$$

No apparent extreme outliers were identified in the numerical variables. Therefore, no additional observations were removed during outlier assessment.

2.2.9 Final Data Quality Check

A final quality check confirmed that:

- the cleaned dataset contained 999 observations and eight variables;
- no missing values remained;
- no duplicate observations remained;
- all numerical variables had appropriate data types; and
- crop-type labels were consistently formatted.

The cleaned dataset was saved as:

`data/processed/crop_yield_cleaned.csv`

This cleaned dataset was used for exploratory analysis, feature engineering and model development.

2.3 Feature Engineering

Feature engineering was carried out to represent possible nonlinear relationships, nutrient composition and interactions among the original predictors. Five additional variables were created from the cleaned dataset.

2.3.1 Total NPK

The total recorded macronutrient level was calculated as the sum of nitrogen, phosphorus and potassium:

$$\text{Total NPK} = N + P + K.$$

This feature summarises the combined level of the three recorded macronutrients.

2.3.2 Nitrogen Proportion

The proportion of total NPK contributed by nitrogen was calculated as:

$$\text{Nitrogen proportion} = \frac{N}{N + P + K}.$$

This feature describes nutrient composition rather than only the absolute nitrogen value.

2.3.3 Phosphorus Proportion

The proportion of total NPK contributed by phosphorus was calculated as:

$$\text{Phosphorus proportion} = \frac{P}{N + P + K}.$$

This feature was included to determine whether the balance of phosphorus relative to the total nutrient level provided additional predictive information.

2.3.4 Temperature Squared

A squared temperature term was created as:

$$\text{Temperature squared} = \text{Temperature}^2.$$

Exploratory analysis suggested that the relationship between temperature and crop yield was mildly nonlinear. The squared term allowed the models to represent possible curvature in this relationship.

2.3.5 Rainfall–Fertilizer Interaction

An interaction term between rainfall and fertilizer application was calculated as:

$$\text{Rainfall–fertilizer interaction} = \text{Rainfall} \times \text{Fertilizer}.$$

This feature was included to represent the possibility that the relationship between fertilizer application and crop yield may vary under different rainfall conditions.

2.3.6 Summary of Engineered Features

Table 1 summarises the engineered variables and their intended interpretation.

Table 1: Summary of engineered features

Feature	**Formula**	**Purpose**
Total NPK	$N + P + K$	Represents the combined recorded level of nitrogen, phosphorus and potassium.
Nitrogen proportion	$\frac{N}{N + P + K}$	Represents the contribution of nitrogen to the total recorded NPK level.
Phosphorus proportion	$\frac{P}{N + P + K}$	Represents the contribution of phosphorus to the total recorded NPK level.
Temperature squared	Temperature^2	Allows the models to capture a possible nonlinear temperature relationship.
Rainfall–fertilizer interaction	Rainfall $\times$ Fertilizer	Allows the models to examine whether fertilizer behaviour varies with rainfall.

The feature-engineering procedure was implemented as a reusable Python function in:

`src/feature_engineering.py`

Using a single reusable function ensured that the engineered variables were calculated consistently during data preparation, model training and Streamlit prediction.

The resulting engineered dataset contained:

- seven original predictor variables;
- five engineered predictor variables;
- one target variable, yield; and
- 999 observations.

The final engineered dataset therefore contained 999 observations and 13 columns. It was saved as:

```
data/processed/crop_yield_engineered.csv
```

Quality checks confirmed that the engineered variables contained no missing or infinite values. The saved dataset was also reloaded and compared with the generated DataFrame to verify that the feature-engineering process had been completed correctly.

The original and engineered feature sets were both retained so that their predictive performances could be compared during model development.

2.4 Data Splitting and Experimental Design

The cleaned dataset contained 999 observations. The target variable was crop yield, while the remaining variables were used as predictors.

Two predictor sets were evaluated:

1. **Original feature set**: crop type, rainfall, temperature, fertilizer, nitrogen, phosphorus and potassium.
2. **Engineered feature set**: the seven original predictors together with total NPK, nitrogen proportion, phosphorus proportion, temperature squared and the rainfall–fertilizer interaction.

For each experiment, the dataset was divided into training and testing subsets using an 80:20 split:

- 799 observations were used for model training;
- 200 observations were reserved for final testing; and
- a random state of 42 was used to make the split reproducible.

The split was implemented using:

```
train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)
```

The same cleaned dataset, row order and random seed were used when comparing the original and engineered feature sets. This ensured that differences in model performance were associated with the feature set or modelling method rather than differences in the test observations.

The test dataset was not used during model fitting or hyperparameter selection. It was retained as an unseen hold-out sample for evaluating the final fitted models.

2.4.1 Reproducibility of the Split

The use of `random_state=42` ensured that the same split could be reproduced when the input dataset and row order remained unchanged.

However, the same random state does not guarantee identical test observations when different versions of the dataset are used. For example, removing or rearranging observations before applying the split can produce a different training and testing composition.

This issue was observed when results produced from the raw 1,000-row dataset were compared with results produced from the cleaned 999-row dataset. Although both workflows used the same random seed, only 84 of the 200 test observations were common to both test sets.

The final analysis therefore used the cleaned 999-row dataset consistently across the evaluated models.

2.4.2 Prevention of Data Leakage

Preprocessing operations required for modelling were included within scikit-learn pipelines. In particular, crop-type encoding was fitted using the training data and then applied to the test data.

This approach prevented information from the test dataset from influencing the fitted preprocessing steps or model parameters.

The engineered variables were created using row-level mathematical transformations based only on values from the same observation. They did not use the target variable or information from other observations. Therefore, their construction did not introduce target leakage.

2.4.3 Experimental Comparison

Four main model experiments were conducted:

1. Multiple Linear Regression using the original feature set;
2. Multiple Linear Regression using the engineered feature set;
3. Random Forest using the original feature set; and
4. Random Forest using the engineered feature set.

The experiments were evaluated using the same performance measures so that the effect of model choice and feature engineering could be compared consistently.

2.5 Consideration of External Data

The project brief permitted the inclusion of additional relevant data from external sources to improve the depth and comprehensiveness of the analysis. Possible external sources considered included weather records, soil databases, satellite information and farm-management data.

However, the supplied dataset did not contain geographic coordinates, location identifiers, dates, production years or farming seasons. These variables would be required to match external observations reliably with the existing crop records.

For example, rainfall or soil data from an external source could only be assigned correctly if the location and time period associated with each observation were known. Adding external data without these identifiers could result in incorrect matches and unsupported assumptions.

External data was therefore not merged with the supplied dataset. This decision was made to preserve the reliability and scientific validity of the analysis rather than introduce information that could not be linked accurately.

The inability to integrate external data was documented as a limitation of the project. Future data collection should include:

- geographic coordinates or regional identifiers;
- planting and harvesting dates;
- production year and farming season;
- soil type and soil pH;
- soil moisture and organic-matter content;
- irrigation practices;
- crop variety;
- pest and disease conditions; and
- farm-management practices.

Including these variables would make it possible to integrate relevant weather, soil and satellite data more reliably in future studies.

3 Exploratory Data Analysis

Exploratory data analysis was conducted to understand the distribution of crop yield, assess differences across crop types, examine relationships between the predictors and yield, and identify patterns that could guide feature engineering and model development.

3.1 Descriptive Statistics

The means and medians of most numerical variables were close, suggesting approximately symmetric distributions with no evidence of severe skewness.

Table 2: Descriptive statistics of the numerical variables

Variable	**Mean**	**Median**	**Minimum**	**Maximum**
Rainfall (mm)	897.57	905.00	400.00	1398.00
Temperature (°C)	29.58	29.00	20.00	40.00
Fertilizer (kg)	73.84	74.00	50.00	99.00
Nitrogen	74.48	75.00	59.00	89.00
Phosphorus	21.82	22.00	15.00	29.00
Potassium	16.93	17.00	10.00	24.00
Yield (Q/acre)	9.86	9.90	5.00	15.00

Rainfall ranged from 400 mm to 1,398 mm, with a mean of 897.57 mm. Temperature ranged from 20°C to 40°C, with a mean of 29.58°C.

Fertilizer application ranged from 50 kg to 99 kg, with an average of 73.84 kg. Nitrogen had the highest recorded nutrient values, followed by phosphorus and potassium.

Crop yield ranged from 5 to 15 quintals per acre. The mean yield was 9.86 quintals per acre, while the median was 9.90 quintals per acre.

The dataset contained six crop categories. The crop categories were reasonably balanced, with the number of observations ranging from 156 for corn to 174 for sorghum.

3.2 Yield Distribution

The crop-yield distribution was approximately symmetric. The mean and median were close, while the skewness value was 0.059.

The small skewness value indicates that the yield distribution did not show substantial left or right skew. The Interquartile Range method also identified no apparent yield outliers.

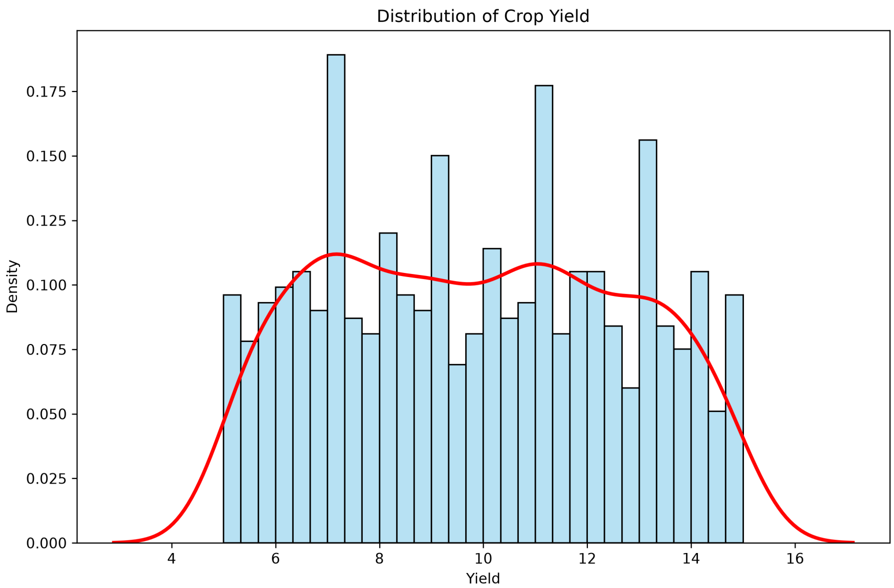

Figure 1: Distribution of crop yield

The absence of severe skewness and extreme outliers meant that no transformation of the target variable was required before modelling.

3.3 Yield by Crop Type

The yield distributions showed substantial overlap across the six crop categories. Although small differences were observed in the mean yields, no crop type had a clearly distinct yield distribution.

A one-way analysis of variance was conducted to determine whether mean crop yield differed significantly across crop types. The result was:

$$F(5, 993) = 0.6494, \qquad p = 0.6621.$$

The result was not statistically significant at the 5% level. Therefore, there was insufficient evidence to conclude that average crop yield differed across the crop categories.

The effect size was:

$$\eta^2 = 0.0033.$$

This means that crop type explained approximately 0.33% of the total variation in crop yield.

Levene's test was also conducted to assess whether yield variance differed across crop types. The result was:

$$W = 0.7264, \qquad p = 0.6037.$$

The non-significant result provided no evidence that the crop groups had unequal yield variances.

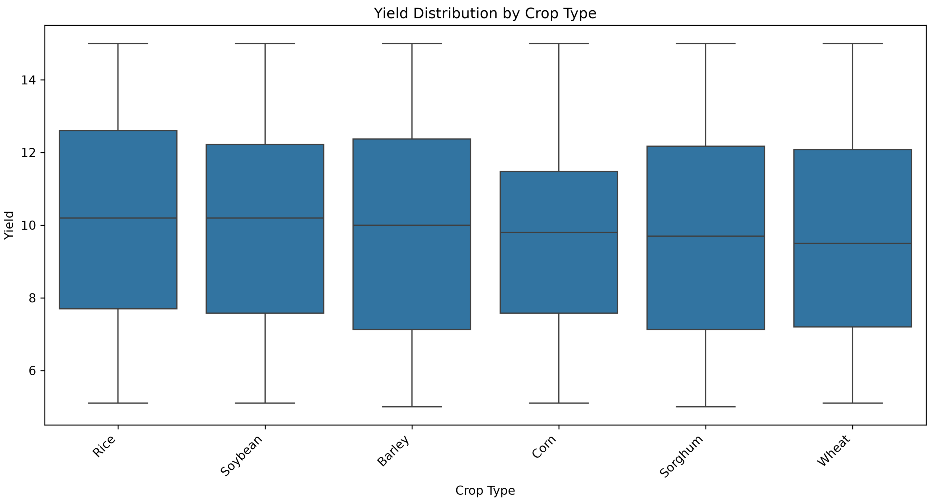

Figure 2: Distribution of crop yield across crop types

Overall, crop type showed only a weak relationship with yield in the available dataset.

3.4 Relationships Between Predictors and Yield

3.4.1 Rainfall and Yield

Rainfall showed a statistically significant but weak positive relationship with crop yield.

Table 3: Association between rainfall and crop yield

Method	**Correlation**	**p-value**
Pearson	0.1027	0.0012
Spearman	0.1065	0.0007

Although the relationships were statistically significant, their magnitudes were small. The squared Pearson correlation was approximately 0.0105, meaning that rainfall alone explained about 1.05% of the variation in crop yield.

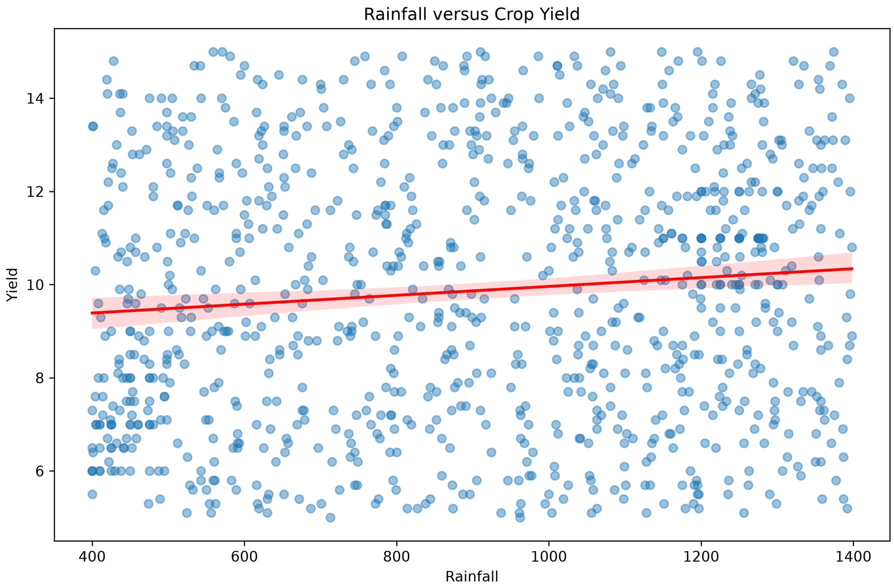

Figure 3: Relationship between rainfall and crop yield

The positive association suggests that yield tended to increase slightly as rainfall increased. However, rainfall was not a strong standalone predictor.

3.4.2 Temperature and Yield

Temperature showed a weak negative relationship with crop yield. The smoothed relationship suggested mild nonlinearity, especially at higher temperature values.

A linear temperature model and a quadratic temperature model were compared.

Table 4: Comparison of linear and quadratic temperature models

Statistic	Linear model	Quadratic model
R^2	0.0171	0.0225
AIC	4881.16	4877.57

The quadratic model produced a slightly higher R^2 and a lower Akaike Information Criterion. The squared temperature term was statistically significant:

$$p = 0.0183.$$

This supported the creation of a squared temperature feature. However, the quadratic model still explained only approximately 2.25% of the variation in yield.

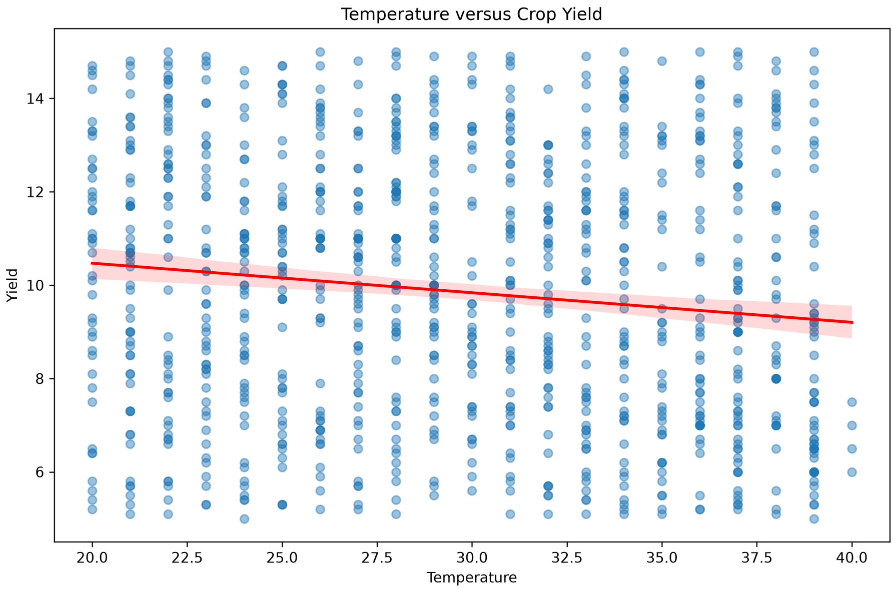

Figure 4: Relationship between temperature and crop yield

3.4.3 Fertilizer and Yield

Fertilizer application showed a very weak positive relationship with crop yield.

Table 5: Association between fertilizer application and crop yield

Method	Correlation	p-value
Pearson	0.0527	0.0960
Spearman	0.0530	0.0944

Neither relationship was statistically significant at the 5% level. Therefore, fertilizer did not show a strong standalone association with yield.

However, the effect of fertilizer may depend on rainfall conditions. This motivated the creation of the rainfall–fertilizer interaction feature.

3.4.4 Macronutrients and Yield

Nitrogen, phosphorus and potassium showed negligible relationships with crop yield.

Table 6: Associations between macronutrients and crop yield

Variable	Pearson	Pearson p-value	Spearman	Spearman p-value
Nitrogen	0.0127	0.6884	0.0112	0.7247
Phosphorus	0.0221	0.4858	0.0244	0.4419
Potassium	-0.0022	0.9441	-0.0006	0.9842

None of the macronutrient variables had a statistically significant standalone association with crop yield.

These findings motivated the creation of total NPK and nutrient-proportion features to determine whether nutrient composition was more informative than the individual nutrient values.

3.5 Outlier Assessment

Box plots and the Interquartile Range method were used to assess potential outliers in rainfall, temperature, fertilizer, nitrogen, phosphorus, potassium and crop yield.

No apparent extreme outliers were identified. Therefore, no additional observations were removed during the exploratory analysis.

3.6 Key EDA Findings

The main findings from the exploratory analysis were:

1. Crop yield was approximately symmetrically distributed and contained no apparent extreme outliers.
2. The crop categories were reasonably balanced.
3. Mean crop yield did not differ significantly across crop types.
4. Rainfall had a statistically significant but weak positive relationship with crop yield.
5. Temperature showed a weak negative and mildly nonlinear relationship with crop yield.
6. Fertilizer showed a weak and statistically non-significant standalone relationship with crop yield.
7. Nitrogen, phosphorus and potassium showed negligible standalone associations with crop yield.
8. The available predictors individually explained only a small proportion of crop-yield variation.
9. The weak relationships observed during exploratory analysis suggested that strong predictive performance might be difficult to achieve using only the supplied variables.

These findings guided the feature-engineering process and provided an important basis for interpreting the later model results.

4 Model Development

The study evaluated three regression model families for crop-yield prediction:

1. Multiple Linear Regression, which served as the baseline model;
2. Random Forest Regression, which was used to capture possible nonlinear relationships and interactions; and

3. Extreme Gradient Boosting, which was evaluated as an additional sequential tree-based ensemble model.

Multiple Linear Regression and Random Forest were each evaluated using both the original and engineered predictor sets. XGBoost was evaluated using the engineered predictor set. This produced five modelling experiments:

1. Multiple Linear Regression with original predictors;
2. Multiple Linear Regression with engineered predictors;
3. Random Forest with original predictors;
4. Random Forest with engineered predictors; and
5. XGBoost with engineered predictors.

4.1 Predictor Sets

The original predictor set contained:

- crop type;
- rainfall;
- temperature;
- fertilizer;
- nitrogen;
- phosphorus; and
- potassium.

The following five engineered variables were created:

- total NPK;
- nitrogen proportion;
- phosphorus proportion;
- temperature squared; and
- rainfall–fertilizer interaction.

For the engineered Multiple Linear Regression and Random Forest models, the five engineered variables were added to the seven original predictors. Each of these models therefore used 12 predictors.

The engineered XGBoost model used crop type, rainfall, temperature, fertilizer and the five engineered variables, giving a total of nine predictors. The individual nitrogen, phosphorus and potassium variables were not included separately in the XGBoost predictor set.

The target variable for all experiments was crop yield, measured in quintals per acre.

4.2 Train-Test Split

The cleaned dataset was divided into training and testing sets before model fitting. Eighty per cent of the observations were assigned to the training set, while the remaining 20% were reserved for final hold-out evaluation.

A random state of 42 was used to make the split reproducible. The same split convention was maintained across the modelling experiments to support a fair comparison of their test-set performance.

4.3 Multiple Linear Regression Baseline

Multiple Linear Regression was used as the baseline model because it provides a simple and interpretable method for examining the combined linear relationship between the predictors and crop yield.

The general multiple linear regression model can be written as:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip} + \varepsilon_i,$$

where:

- Y_i represents crop yield for observation i;
- β_0 is the intercept;
- $\beta_1, \ldots, \beta_p$ are the regression coefficients;
- $X_{i1}, \ldots, X_{ip}$ are the predictor values; and
- ε_i is the unexplained error term.

The model assumes that crop yield can be represented as a linear combination of the predictors.

Two linear regression models were fitted:

1. a baseline model using only the original predictors; and
2. an engineered model using the original and engineered predictors.

The engineered linear regression model was used to determine whether the additional nonlinear, interaction and nutrient-composition variables improved predictive performance.

4.3.1 Role of the Baseline Model

The linear regression models provided reference points for evaluating the more flexible tree-based models. If Random Forest or XGBoost produced substantially better results, this would suggest that nonlinear relationships or interactions were important.

However, if all models performed similarly, this would suggest that limited model performance was more likely related to weak predictive information in the available data than to the assumptions of the linear model.

4.4 Random Forest Regressor

Random Forest is an ensemble learning method that combines predictions from multiple decision trees.

Each tree is trained using a bootstrap sample of the training data. At each split, the tree considers a random subset of the available predictors. These two sources of randomness help create diverse trees.

For regression, the final Random Forest prediction is the average prediction from all individual trees:

$$\widehat{Y} = \frac{1}{B}\sum_{b=1}^{B}\widehat{Y}_b,$$

where:

- B is the total number of decision trees; and
- $\widehat{Y}_b$ is the prediction produced by tree b.

Averaging across multiple trees generally reduces the instability and variance associated with a single decision tree.

Random Forest was selected because it can:

- represent nonlinear relationships;
- capture interactions among variables;
- handle numerical and encoded categorical predictors;
- reduce the overfitting associated with individual decision trees; and
- provide feature-importance information for model interpretation.

Two Random Forest models were fitted:

1. Random Forest using the original predictor set; and
2. Random Forest using the original and engineered predictor set.

4.5 Extreme Gradient Boosting

Extreme Gradient Boosting, commonly known as XGBoost, was evaluated as an additional nonlinear ensemble model.

Unlike Random Forest, which constructs trees independently and averages their predictions, XGBoost constructs trees sequentially. Each new tree attempts to reduce the prediction errors remaining from the existing ensemble.

The engineered XGBoost predictor set contained:

- crop type;
- rainfall;
- temperature;

- fertilizer;
- total NPK;
- nitrogen proportion;
- phosphorus proportion;
- temperature squared; and
- rainfall–fertilizer interaction.

XGBoost was included to determine whether a sequential boosting method could provide meaningful predictive improvement over Multiple Linear Regression and Random Forest.

4.6 Preprocessing Pipelines

Scikit-learn pipelines were used to combine preprocessing and model fitting for the Random Forest and XGBoost models.

The categorical predictor, crop type, was converted into numerical indicator variables using one-hot encoding. The encoder was configured to ignore previously unseen categories during prediction.

The numerical predictors were passed directly to the tree-based models without scaling. Feature scaling was not required because Random Forest and XGBoost make predictions using threshold-based decision-tree splits rather than distance calculations.

The general preprocessing structure can be summarised as:

$$\text{Input data} \longrightarrow \text{Categorical encoding} \longrightarrow \text{Numerical passthrough} \longrightarrow \text{Regression model.}$$

Combining preprocessing and modelling within a pipeline provided several advantages:

- preprocessing was fitted using the training data;
- the same transformations were applied consistently to validation and test observations;
- the risk of data leakage was reduced;
- hyperparameter tuning could be performed on the complete modelling workflow; and
- the selected Random Forest pipeline could be saved and used in the Streamlit application.

4.7 Hyperparameter Tuning

Hyperparameter tuning was conducted for the Random Forest and XGBoost models using `GridSearchCV`. Each parameter combination was assessed using five-fold cross-validation on the training dataset.

The scoring measure used during tuning was negative root mean squared error. Scikit-learn represents error measures as negative values during optimisation so that larger scores correspond to better performance. The parameter combination with the lowest cross-validated RMSE was retained.

Each selected model was subsequently refitted using the complete training dataset before evaluation on the unseen test dataset.

4.7.1 Random Forest Hyperparameter Search

The Random Forest hyperparameters controlled the number, complexity and diversity of the decision trees.

The parameters considered were:

- `n_estimators`: the number of trees in the forest;
- `max_depth`: the maximum depth allowed for each tree;
- `min_samples_leaf`: the minimum number of observations required in a terminal leaf; and
- `max_features`: the number of predictors considered when selecting each split.

The Random Forest search space is shown in Table 7.

Table 7: Random Forest hyperparameter search space

Hyperparameter	**Values evaluated**
Number of trees	100, 200
Maximum tree depth	5, 10, unrestricted
Minimum samples per leaf	1, 2
Features considered per split	Square root

The grid search selected the following hyperparameters for the engineered Random Forest model:

- number of trees: 200;
- maximum tree depth: 5;
- minimum number of observations per leaf: 2; and
- number of features considered at each split: square root.

The best cross-validated RMSE obtained during Random Forest hyperparameter tuning was 2.7079. The selected parameter combination was subsequently refitted using the complete training set.

4.7.2 XGBoost Hyperparameter Search

The XGBoost search considered the number and depth of the trees, the learning rate, row and column subsampling, and L2 regularisation.

The XGBoost search space is shown in Table 8.

Table 8: XGBoost hyperparameter search space

Hyperparameter	**Values evaluated**
Number of estimators	100, 200, 300
Maximum tree depth	3, 5, 7
Learning rate	0.01, 0.05, 0.10
Subsample ratio	0.7, 1.0
Column-sampling ratio	0.7, 1.0
L2 regularisation parameter	1, 5

The selected XGBoost hyperparameters were:

- number of estimators: 100;
- maximum tree depth: 3;
- learning rate: 0.01;
- subsample ratio: 1.0;
- column-sampling ratio: 1.0; and
- L2 regularisation parameter: 1.

The best cross-validated RMSE obtained during XGBoost hyperparameter tuning was 2.7212. The selected parameter combination was subsequently refitted using the complete training set.

4.8 Cross-Validation Strategy

A single train-test split provides only one estimate of model performance and may be affected by the particular observations assigned to the test set. Five-fold cross-validation was therefore used for hyperparameter selection and final model assessment.

Hyperparameter tuning was conducted using only the training dataset. During the Random Forest grid search, five-fold cross-validation was specified through `cv=5`. For XGBoost tuning, the folds were shuffled, and a random state of 42 was used to make the procedure reproducible.

After hyperparameter selection, the fitted models were also evaluated using five-fold cross-validation on the complete modelling dataset. For this final assessment, the observations were shuffled before splitting, and a random state of 42 was used.

During each cross-validation iteration:

1. four folds were used to fit the model;
2. the remaining fold was used for validation;

3. a different fold was used for validation during the next iteration; and
4. the process continued until every fold had been used once for validation.

The fold-level RMSE values were averaged to obtain the mean cross-validation RMSE. The corresponding standard deviation was also reported to describe the variability of model performance across the five folds. A smaller standard deviation indicates more consistent performance across different subsets of the data.

4.9 Evaluation Metrics

The models were evaluated using Mean Absolute Error, Root Mean Squared Error and the coefficient of determination, R^2.

4.9.1 Mean Absolute Error

Mean Absolute Error measures the average absolute difference between the observed and predicted yields:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \widehat{y}_i\right|.$$

MAE is expressed in the same unit as crop yield. A lower MAE indicates better predictive accuracy.

For example, an MAE of 2.5 quintals per acre means that the model predictions differ from the observed yields by approximately 2.5 quintals per acre on average.

4.9.2 Root Mean Squared Error

Root Mean Squared Error is calculated as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}.$$

RMSE gives greater weight to large prediction errors because the errors are squared before averaging. A lower RMSE indicates better predictive performance.

4.9.3 Coefficient of Determination

The coefficient of determination is calculated as:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}\left(y_i - \widehat{y}_i\right)^2}{\sum_{i=1}^{n}\left(y_i - \overline{y}\right)^2}.$$

R^2 measures the proportion of variation in crop yield explained by the model.

Its interpretation is:

- an R^2 close to 1 indicates that the model explains a large proportion of yield variation;

- an R^2 close to 0 indicates that the model performs similarly to predicting the mean yield; and
- a negative R^2 indicates that the model performs worse than predicting the mean yield for every observation.

4.10 Model Comparison Procedure

The five modelling experiments were compared using:

- hold-out test MAE;
- hold-out test RMSE;
- hold-out test R^2;
- mean five-fold cross-validation RMSE;
- cross-validation variability; and
- the difference between training and testing performance.

The train-test performance gap was examined to identify possible overfitting. A model that performs substantially better on the training data than on the test data may have learned patterns that do not generalise to unseen observations.

Final model selection was based on predictive performance, stability, interpretability and deployment considerations. The model with the smallest single numerical error was not automatically considered the most reliable.

Model performance was interpreted alongside the exploratory findings, cross-validation results and limitations of the available data.

5 Results and Interpretation

This section presents the predictive performance of the Multiple Linear Regression, Random Forest and XGBoost models. Five modelling experiments were compared using hold-out test performance and five-fold cross-validation.

The results are interpreted alongside the exploratory findings. Since the exploratory analysis revealed generally weak relationships between the available predictors and crop yield, limited predictive performance was expected.

5.1 Model Comparison

Table 9 summarises the performance of the five evaluated models.

Table 9: Comparison of the evaluated crop-yield prediction models

Model	**Feature set**	**Test MAE**	**Test RMSE**	**Test R^2**	**CV RMSE**
Multiple Linear Regression	Original	2.5342	2.9391	-0.0026	2.7926
Multiple Linear Regression	Engineered	2.5525	2.9453	-0.0068	2.7892
Random Forest	Original	2.5460	2.9390	-0.0030	2.7620 ± 0.1360
Random Forest	Engineered	**2.5250**	2.9260	0.0060	$\mathbf{2.7620 \pm 0.1250}$
XGBoost	Engineered	2.5395	**2.9219**	**0.0091**	2.7699 ± 0.1183

The engineered Random Forest achieved the lowest test MAE and the lowest mean cross-validation RMSE. The engineered XGBoost model achieved the lowest hold-out test RMSE and the highest test R^2.

However, the differences between the two ensemble models were extremely small. XGBoost reduced test RMSE by only 0.0041 quintals per acre compared with the engineered Random Forest. Random Forest reduced test MAE by 0.0145 quintals per acre and achieved a slightly lower mean cross-validation RMSE.

The highest test R^2 was only 0.0091. This means that even the strongest hold-out result explained less than 1% of the observed variation in unseen crop yields.

Therefore, none of the evaluated models demonstrated strong predictive performance.

5.2 Multiple Linear Regression Results

5.2.1 Original Feature Set

The Multiple Linear Regression model trained using the original predictors achieved:

- test MAE of 2.5342 quintals per acre;
- test RMSE of 2.9391 quintals per acre;
- test R^2 of -0.0026;
- mean cross-validation MAE of 2.3841;
- mean cross-validation RMSE of 2.7926; and
- mean cross-validation R^2 of 0.0014.

The slightly negative test R^2 indicates that the model performed marginally worse than predicting the mean test-set yield for every observation.

The cross-validation R^2 was close to zero, showing that the original predictors explained very little variation in crop yield under a linear modelling assumption.

5.2.2 Engineered Feature Set

The engineered Multiple Linear Regression model achieved:

- test MAE of 2.5525 quintals per acre;
- test RMSE of 2.9453 quintals per acre;

- test R^2 of -0.0068;
- mean cross-validation MAE of 2.3780;
- mean cross-validation RMSE of 2.7892; and
- mean cross-validation R^2 of 0.0039.

The engineered model produced a slightly lower mean cross-validation RMSE than the original linear model. However, it performed marginally worse on the hold-out test set.

The differences were too small to represent a meaningful improvement. The original-feature model was therefore retained as the simpler linear baseline.

5.2.3 Interpretation of the Linear Results

The weak performance of both linear models was consistent with the exploratory analysis.

Rainfall, temperature, fertilizer and the nutrient variables showed weak standalone relationships with crop yield. Adding mathematically derived variables did not substantially increase the predictive information available to the linear model.

The results suggest that crop yield could not be explained adequately as a linear combination of the available predictors.

5.3 Random Forest Results

5.3.1 Original Feature Set

The Random Forest trained using the original predictor set achieved the results shown in Table 10.

Table 10: Performance of the original-feature Random Forest

Metric	Training set	Test set
MAE	2.113	2.546
RMSE	2.491	2.939
R^2	0.190	-0.003

The five-fold cross-validation RMSE was:

$$2.762 \pm 0.136.$$

The model explained approximately 19.0% of the variation in the training data but failed to explain meaningful variation in the test data.

The difference between the training and testing results indicates that the model learned some patterns from the training data that did not generalise effectively to unseen observations.

5.3.2 Engineered Feature Set

The engineered Random Forest achieved the results shown in Table 11.

Table 11: Performance of the engineered-feature Random Forest

Metric	**Training set**	**Test set**
MAE	2.093	2.525
RMSE	2.480	2.926
R^2	0.197	0.006

The five-fold cross-validation RMSE was:

$$2.762 \pm 0.125.$$

Compared with the original-feature Random Forest:

- test MAE decreased from 2.546 to 2.525;
- test RMSE decreased from 2.939 to 2.926;
- test R^2 increased from -0.003 to 0.006;
- mean cross-validation RMSE remained approximately 2.762; and
- cross-validation variability decreased from 0.136 to 0.125.

These differences were numerically favourable but practically negligible.

The engineered variables therefore produced only a very small improvement in hold-out performance and did not substantially improve model generalisation.

5.4 XGBoost Results

The engineered XGBoost model achieved the results shown in Table 12.

Table 12: Performance of the engineered-feature XGBoost model

Metric	**Training set**	**Test set**
MAE	2.2411	2.5395
RMSE	2.6322	2.9219
R^2	0.0957	0.0091

The five-fold cross-validation RMSE was:

$$2.7699 \pm 0.1183.$$

The XGBoost model achieved the lowest hold-out test RMSE and the highest test R^2 among the five modelling experiments.

However, the test R^2 of 0.0091 remained close to zero, showing that the model explained less than 1% of the variation in unseen crop yields.

The difference between the XGBoost training and testing errors was smaller than the corresponding difference for the Random Forest models. This indicates that the selected XGBoost configuration was relatively conservative and did not severely overfit the training dataset.

Nevertheless, its overall predictive performance remained weak. The improvement in hold-out RMSE over the engineered Random Forest was only 0.0041 quintals per acre and was not practically meaningful.

5.5 Prediction Performance

Actual-versus-predicted plots were used to examine how closely the engineered Random Forest and XGBoost predictions followed the observed crop yields.

5.5.1 Engineered Random Forest Predictions

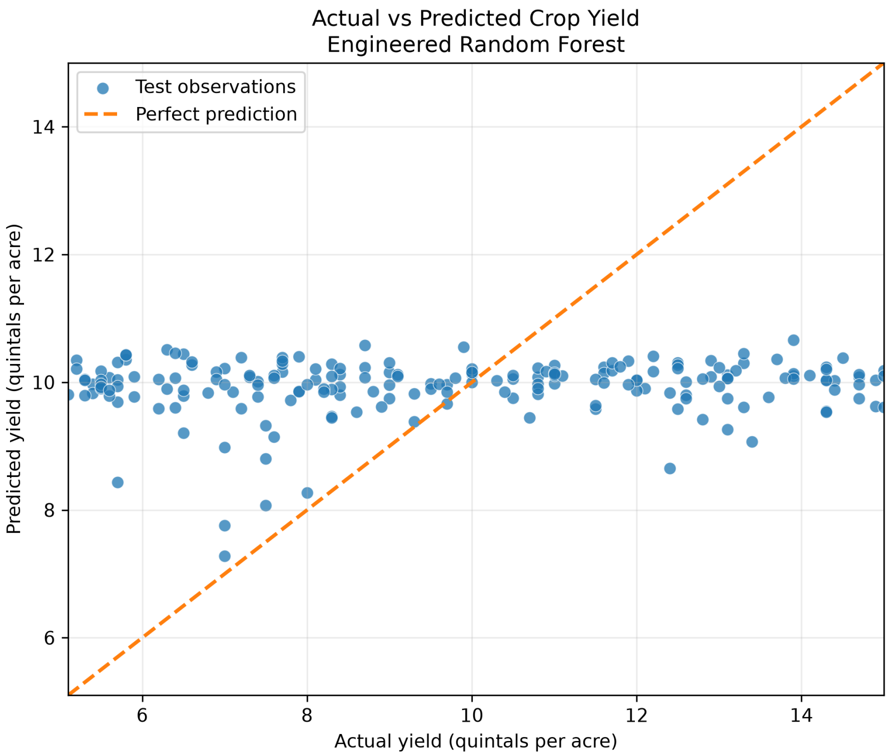

Figure 5: Actual and predicted crop yields for the engineered Random Forest

A well-performing regression model would produce observations located close to the 45-degree reference line.

Figure 5 shows that the Random Forest predictions were concentrated around the central yield range. The model did not closely reproduce the observed variation between low-yield and high-yield observations.

The model therefore tended to predict values close to the average yield. This pattern was consistent with its test R^2 of 0.006.

5.5.2 Engineered XGBoost Predictions

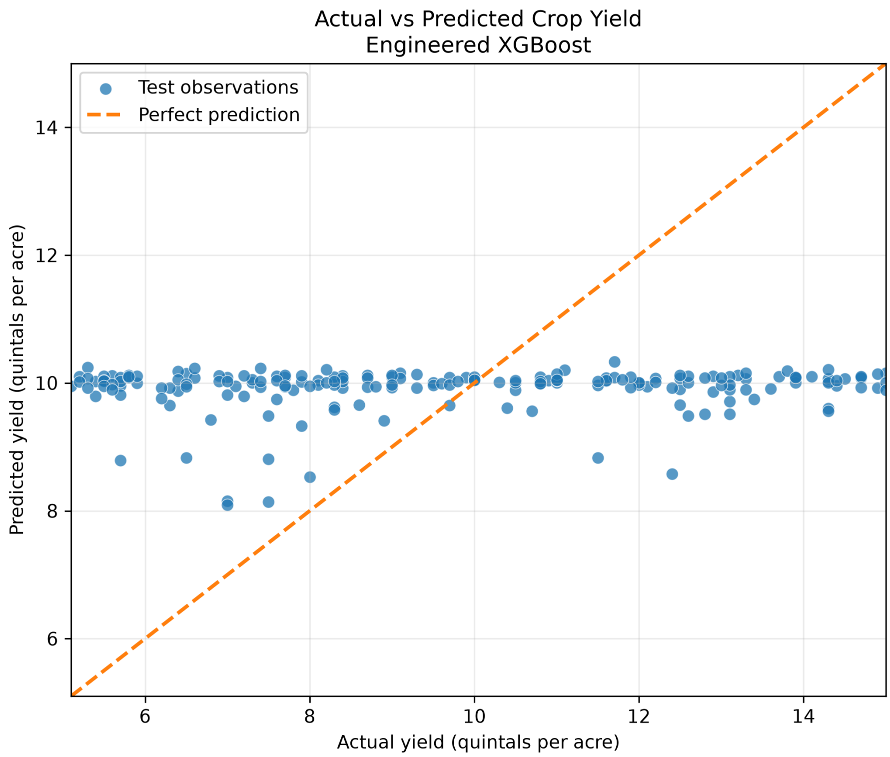

Figure 6: Actual and predicted crop yields for the engineered XGBoost model

Figure 6 shows a similar concentration of predictions around the central crop-yield range.

Although XGBoost produced a marginally lower hold-out RMSE than Random Forest, it also had difficulty distinguishing observations with unusually low or high crop yields. This was consistent with its test R^2 of 0.0091.

5.6 Impact Analysis of the Selected Random Forest

The impact analysis was conducted only for the engineered Random Forest because it was retained as the final deployment model. XGBoost was evaluated for predictive comparison but was not subjected to the full interpretation workflow.

The project brief required an assessment of the variables that most strongly influenced crop-yield prediction.

Four complementary interpretation methods were used:

1. impurity-based feature importance;
2. permutation importance;
3. partial dependence analysis; and
4. crop-type scenario analysis.

These methods describe the behaviour of the fitted Random Forest model. They do not establish that the identified variables have causal effects on actual crop yield.

The results were interpreted cautiously because the selected model demonstrated limited predictive performance on unseen observations.

5.6.1 Impurity-Based Feature Importance

Impurity-based feature importance measures the total reduction in prediction error produced by each feature across all decision-tree splits.

A feature receives a higher importance score when it is used frequently to create splits that reduce variation within the resulting groups.

The method provides a convenient summary of how the trees were constructed. However, it may favour continuous variables or variables with many possible split points.

For this reason, impurity-based importance was treated as a supplementary interpretation method rather than the main measure of predictive contribution.

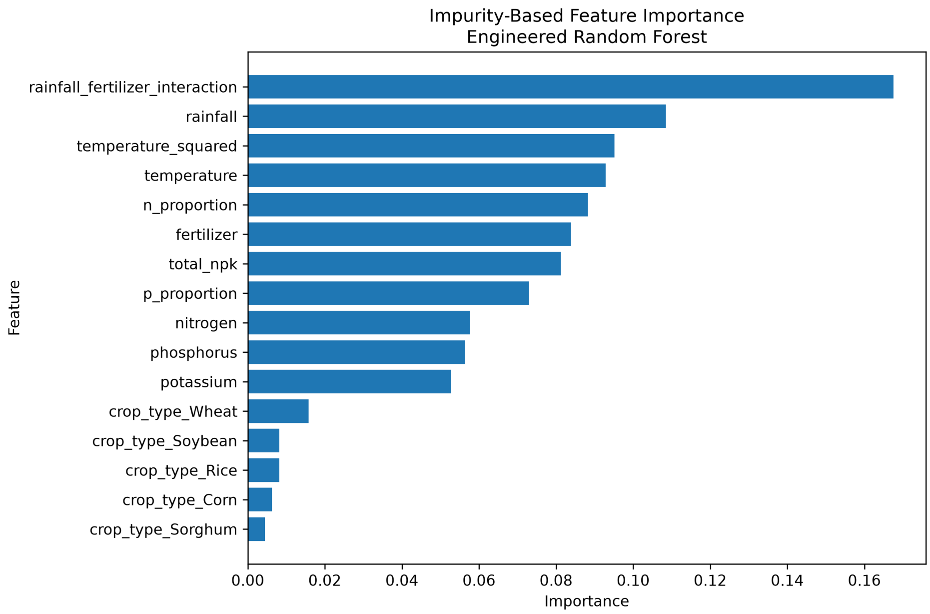

Figure 7: Impurity-based feature importance for the engineered Random Forest

5.6.2 Permutation Importance

Permutation importance measures how much model performance worsens when the values of one feature are randomly shuffled.

Shuffling removes the relationship between that feature and the target while leaving the remaining predictors unchanged.

A feature is considered more useful when shuffling it produces a larger increase in prediction error.

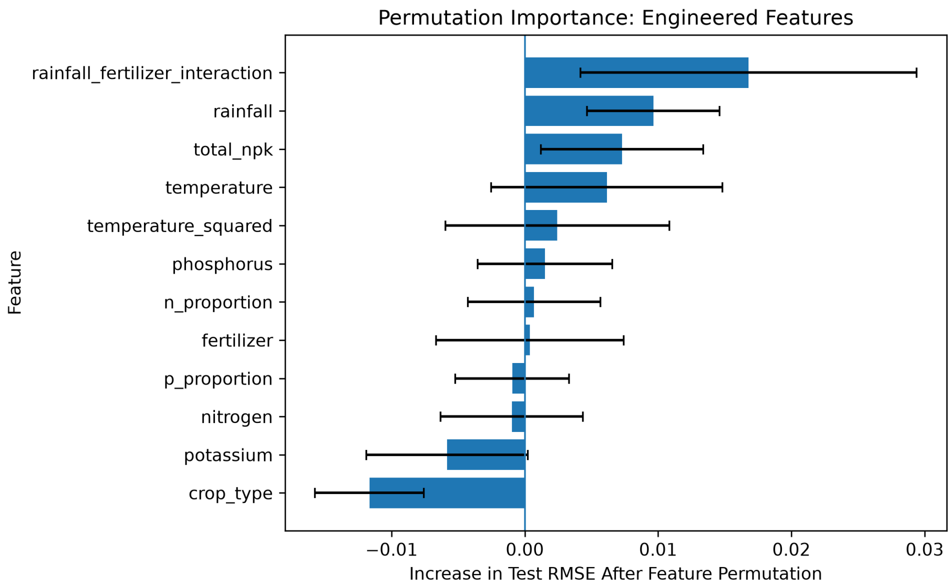

Figure 8: Permutation importance for the engineered Random Forest

Rainfall and temperature produced the largest positive permutation-importance values among the available predictors. This suggests that the fitted model relied on them more than on the remaining features.

Phosphorus showed a small but uncertain contribution, while nitrogen and fertilizer had importance values close to zero.

Potassium and crop type produced negative mean importance values. This suggests that they did not improve test-set prediction and may have contributed noise or instability.

However, the importance values were small, and the uncertainty intervals for several features crossed or approached zero. Rainfall and temperature were therefore the most useful variables only in a relative sense. They were not strong predictors in absolute terms.

Permutation importance may also be divided among related variables. For example, temperature and temperature squared contain overlapping information. Similarly, nitrogen, phosphorus and potassium are mathematically related to total NPK and the nutrient proportions.

5.6.3 Partial Dependence Analysis

Partial dependence plots were used to examine how the model's average predicted yield changed as a numerical feature varied.

For each selected value of a feature, the model generated predictions while the remaining variables were retained at their observed values. The predictions were then averaged across the observations.

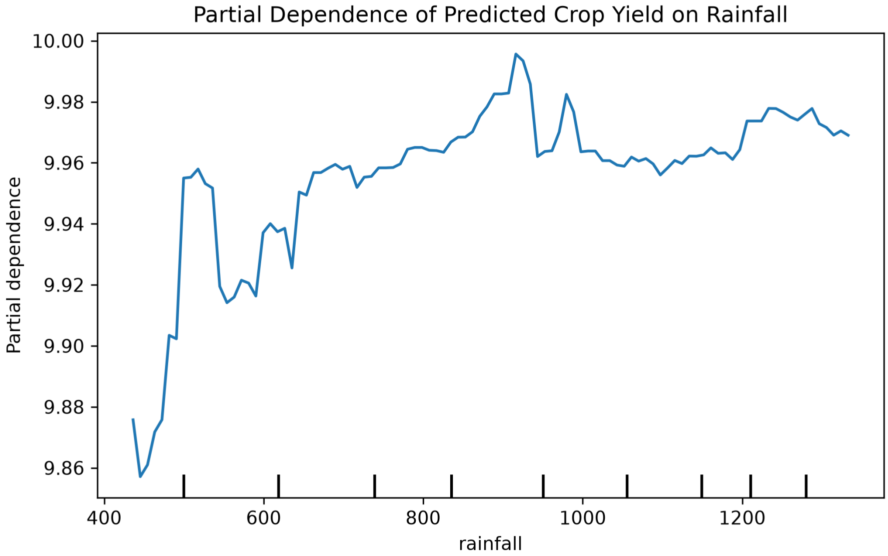

Figure 9: Partial dependence of predicted crop yield on rainfall

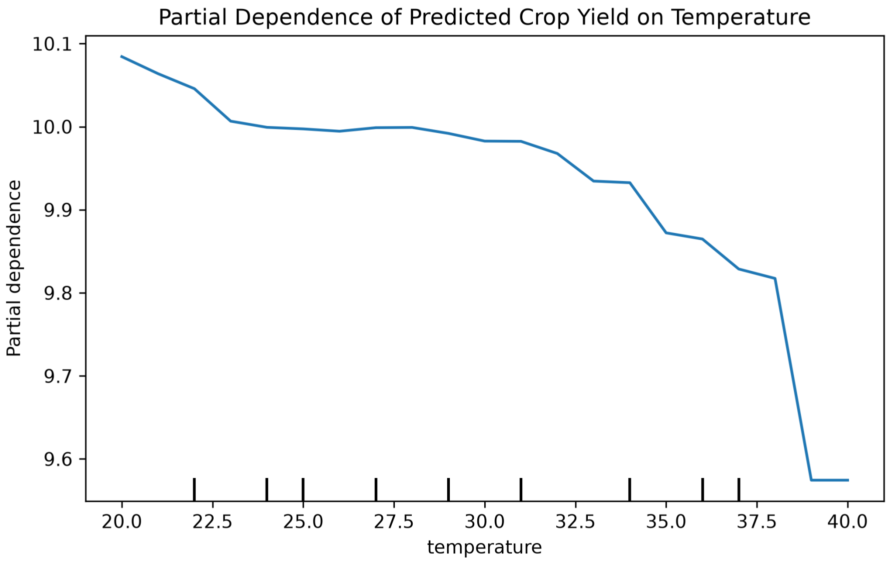

Figure 10: Partial dependence of predicted crop yield on temperature

The partial dependence plots showed only modest changes in average predicted yield across the observed predictor ranges.

Rainfall displayed a slight positive pattern, while temperature showed mild nonlinear variation. However, neither pattern was strong.

These findings were consistent with the exploratory analysis, which found weak relationships between rainfall, temperature and crop yield.

The plots represent average relationships learned by the model. They do not imply that directly changing rainfall or temperature would cause an equivalent change in actual crop yield.

5.6.4 Crop-Type Impact Analysis

Crop-type impact was assessed by assigning each crop category to all test observations in turn while keeping the remaining variables unchanged.

The average predicted yield was then calculated for each crop category.

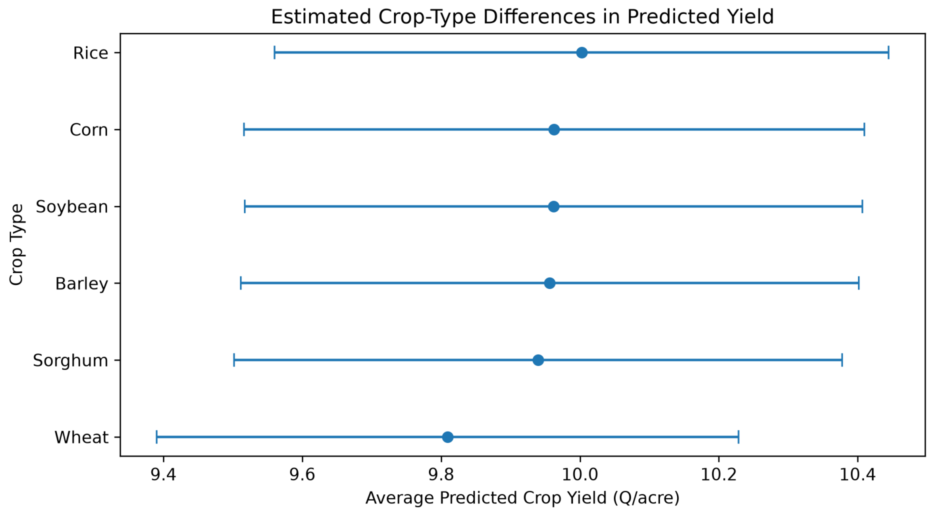

Figure 11: Average model prediction under each crop-type scenario

The average predicted yields differed only slightly across crop categories.

This result agreed with the exploratory analysis, where the analysis of variance found no statistically significant difference in mean yield across crop types and the effect size was negligible.

The crop-type analysis describes differences in model predictions. It should not be interpreted as evidence that changing crop type would cause yield to increase or decrease.

5.6.5 Summary of Impact Analysis

The four interpretation methods answered different questions:

- **Impurity-based importance** identified the variables most frequently used by the decision trees to reduce prediction error.
- **Permutation importance** assessed which variables were most useful for maintaining predictive performance.
- **Partial dependence plots** showed how average predictions changed as numerical variables changed.
- **Crop-type scenario analysis** showed how average model predictions differed across crop categories.

Overall, rainfall and temperature were the most useful original predictors relative to the other available variables.

However, their absolute predictive contribution remained small. The impact-analysis results therefore explain the behaviour of the fitted model but do not provide strong evidence of real-world agronomic effects.

5.7 Final Model Selection

The engineered Random Forest was retained as the final deployment model.

The engineered XGBoost model recorded the lowest hold-out test RMSE and the highest test R^2. However, its improvement over the engineered Random Forest was extremely small. XGBoost reduced test RMSE by only 0.0041 quintals per acre.

The engineered Random Forest achieved:

- the lowest test MAE;
- a slightly lower mean cross-validation RMSE than XGBoost;
- competitive hold-out RMSE and R^2 values;
- a complete model-interpretation workflow; and
- a fitted preprocessing and prediction pipeline that had already been successfully incorporated into the Streamlit application.

Its final performance was:

- test MAE: 2.525 quintals per acre;
- test RMSE: 2.926 quintals per acre;
- test R^2: 0.006; and
- five-fold cross-validation RMSE: 2.762 ± 0.125.

XGBoost had slightly lower cross-validation variability, with a standard deviation of 0.1183 compared with 0.1250 for the engineered Random Forest. However, Random Forest achieved the better mean cross-validation RMSE and lower test MAE.

Replacing the deployed Random Forest with XGBoost would therefore have added implementation work without producing a meaningful improvement in predictive performance.

The Random Forest was selected as the most suitable model for the technical prototype, not because it achieved high predictive accuracy.

It should not be presented as a reliable production model without further data collection, external validation and model improvement.

5.8 Interpretation of the Overall Results

The results showed that neither the linear nor the nonlinear models extracted substantial predictive information from the supplied variables.

Multiple Linear Regression produced R^2 values close to zero. Random Forest captured more variation in the training data, but this did not translate into meaningful test performance. XGBoost also achieved a test R^2 close to zero.

This suggests that the weak results were not caused only by the linear assumptions of the baseline model or by the use of a particular modelling algorithm.

Feature engineering was supported by exploratory analysis and agricultural reasoning. However, the additional variables were mathematical transformations of the existing predictors. They could not introduce important information that was absent from the original dataset.

The main limitation therefore appears to be the information content of the available variables rather than the choice of algorithm.

Important factors that were not recorded include:

- geographic location;
- production season and year;
- soil type and soil pH;
- soil moisture;
- irrigation;
- crop variety;
- planting and harvesting dates;
- pest and disease conditions; and
- farm-management practices.

The exploratory analysis and model results were consistent with each other. Both indicated that rainfall, temperature, fertilizer, nutrients and crop type explained only a small proportion of the observed crop-yield variation.

The poor predictive performance is therefore an important project finding. It demonstrates the need for more informative and better-linked agricultural data rather than an unsupported attempt to obtain high accuracy through increasingly complex models.

6 Recommendations

The recommendations are based on the exploratory analysis, the performance of the five modelling experiments, the Random Forest impact analysis and the limitations of the available dataset.

The highest test R^2 among the evaluated models was only 0.0091, while the selected Random Forest achieved a test R^2 of 0.006. The recommendations therefore focus primarily on improving data quality, strengthening future model development and defining the appropriate use of the deployed prototype.

6.1 Recommendations for the Firm

The firm should treat the current crop-yield prediction system as a technical prototype rather than as a validated agricultural decision-support tool.

The application demonstrates a complete data-science workflow involving:

- data cleaning and preparation;
- exploratory data analysis;
- feature engineering;

- linear, bagging and boosting models;
- hyperparameter tuning;
- model evaluation and comparison;
- interpretation of the selected model; and
- Streamlit deployment.

However, the current predictions should not be used as the sole basis for production, fertiliser, lending, financing, investment or resource-allocation decisions.

The firm should first improve the quality, scope and documentation of its agricultural data before developing a production-level prediction system.

The firm should also establish a consistent data-governance process. This should include:

- clear definitions for all variables;
- confirmed measurement units;
- consistent data-entry formats;
- unique identifiers for each farm, field or observation;
- geographic and temporal identifiers;
- regular checks for missing values, duplicates and invalid entries; and
- documentation of all data-cleaning, feature-engineering and modelling procedures.

A standardised data-collection process would make future datasets easier to analyse and improve the reliability and reproducibility of subsequent prediction models.

6.2 Recommendations for Future Data Collection

The current dataset contains only a limited set of environmental, fertiliser and nutrient variables. Future data collection should include additional factors that are directly related to crop performance.

Recommended variables include:

- farm location or geographic coordinates;
- district, region or production zone;
- production year and farming season;
- planting and harvesting dates;
- crop variety;
- soil type;
- soil pH;

- soil moisture;
- organic-matter content;
- irrigation practices;
- fertiliser type, quantity and application schedule;
- pest and disease conditions;
- farm-management practices;
- previous crop grown on the land; and
- remotely sensed vegetation, weather or land-condition indicators.

These additional variables may explain crop-yield variation that could not be captured by the current dataset.

The firm should also aim to collect a larger number of observations across different farms, locations, crop varieties, seasons and production years.

A larger and more diverse dataset would improve the ability of machine-learning models to identify stable relationships and generalise to new observations.

6.3 Recommendations for External Data Integration

External data should only be added when they can be matched reliably to the existing observations.

For example, external rainfall, temperature, soil or satellite information should be linked using:

- geographic coordinates;
- farm or location identifiers;
- observation dates;
- production years; or
- farming seasons.

Without these identifiers, external records may be assigned incorrectly to farms or crop observations.

The firm should therefore avoid merging external data based only on broad averages, assumptions or approximate location descriptions.

When reliable identifiers become available, possible external sources may include:

- national meteorological records;
- soil databases;
- satellite-derived vegetation indices;
- gridded rainfall estimates;

- land-cover information; and
- regional agricultural statistics.

Any external dataset should be reviewed for compatibility, measurement units, spatial resolution, temporal resolution, coverage period and data quality before integration.

6.4 Recommendations for Model Improvement

Future modelling work should begin after the dataset has been strengthened with more informative predictors.

Once improved data are available, the firm may consider:

- comparing additional regression algorithms;
- expanding the Random Forest and XGBoost hyperparameter searches;
- using randomised search or Bayesian optimisation;
- examining crop-specific models;
- including spatial and temporal predictors;
- investigating interactions supported by agricultural knowledge;
- testing models across different seasons, regions and production years;
- using repeated or nested cross-validation;
- evaluating uncertainty around individual predictions; and
- comparing all models against simple benchmark methods.

A dummy regressor that predicts the mean crop yield should be included as a formal benchmark. Any advanced model should show a clear, stable and practically meaningful improvement over this benchmark before being considered useful.

Future models should also be evaluated separately by crop type when sufficient observations are available for each category. This may reveal whether certain crops are more predictable than others.

Where suitable geographic and temporal identifiers become available, future work may also investigate spatial, temporal or spatiotemporal modelling approaches.

The current results indicate that greater algorithmic complexity alone is unlikely to produce a major improvement without richer data.

6.5 Recommendations for Model Interpretation

Feature-importance and partial-dependence results should be interpreted as descriptions of model behaviour rather than evidence of causal agricultural effects.

The current Random Forest impact analysis suggested that rainfall and temperature were relatively more useful than the remaining predictors. However, their absolute predictive contributions were small.

The firm should therefore avoid statements such as:

> Increasing rainfall will cause crop yield to increase.

A more appropriate interpretation is:

> Within the fitted Random Forest model, rainfall was one of the variables that contributed relatively more to prediction than the other available predictors.

Similarly, differences in predicted yield across crop-type scenarios should not be presented as evidence that changing crop type would cause yield to increase or decrease.

Causal recommendations would require stronger observational data, experimental evidence or a carefully designed causal analysis.

6.6 Recommendations for Prototype Use

The Streamlit application may be used for:

- demonstrating the end-to-end prediction workflow;
- testing how user inputs are processed;
- showing how engineered features are created automatically;
- presenting the selected model to project stakeholders;
- supporting training or educational demonstrations; and
- providing a foundation for future system development.

The application should continue to display a clear warning that the deployed model has limited predictive performance.

Users should be informed that:

- the prediction is an estimate produced by a weak model;
- the model was trained using a limited dataset;
- several important agricultural variables were not included;
- the model has not been externally validated;
- the prediction does not include an uncertainty interval; and
- the output should not replace expert agricultural advice.

The prototype should only be upgraded to a production decision-support system after:

1. more informative and representative data have been collected;
2. the model has been retrained using the improved dataset;
3. performance has improved clearly and consistently;
4. predictions have been validated using independent agricultural data;
5. appropriate uncertainty estimates have been added;
6. usability testing has been conducted with intended users; and
7. the system has been reviewed by agricultural and domain experts.

6.7 Recommendations for Monitoring and Validation

Any future operational version of the system should include a process for monitoring model performance over time.

The firm should:

- record model inputs and verified crop-yield outcomes;
- compare predictions with actual harvest results;
- monitor changes in data distributions;
- assess whether model accuracy differs across crops, locations or seasons;
- retrain the model when sufficient new data become available; and
- maintain clear version records for datasets, features and deployed models.

This process would help identify model deterioration and determine whether a future model remains suitable for its intended use.

6.8 Summary of Recommendations

The main recommendation is that the firm should prioritise better data collection and documentation rather than relying only on more complex algorithms.

The current analysis showed that feature engineering, Multiple Linear Regression, Random Forest and XGBoost could not overcome the limited predictive information contained in the supplied dataset.

A more reliable crop-yield prediction system will require richer agricultural data, clear observation identifiers, consistent measurement standards, careful external-data integration, independent validation and ongoing model monitoring.

Until these requirements are met, the Streamlit application should remain a technical and educational prototype.

7 Dashboard and Deployment

A Streamlit application was developed to demonstrate how the selected engineered Random Forest model could be used through an interactive interface.

The application allows a user to enter crop, environmental, fertilizer and nutrient information and receive an estimated crop yield in quintals per acre.

The dashboard was designed as a technical prototype. It demonstrates the complete prediction workflow but is not presented as a validated agricultural decision-support system because the selected model achieved limited predictive performance.

7.1 Selected Deployment Model

The Random Forest model trained using the original and engineered predictor sets was retained as the deployment model.

Its final performance was:

- test MAE of 2.525 quintals per acre;

- test RMSE of 2.926 quintals per acre;
- test R^2 of 0.006; and
- five-fold cross-validation RMSE of 2.762 ± 0.125.

The engineered XGBoost model recorded a marginally lower hold-out test RMSE of 2.9219 and a slightly higher test R^2 of 0.0091. However, the improvement in test RMSE over Random Forest was only 0.0041 quintals per acre.

The engineered Random Forest achieved the lower test MAE and a slightly better mean cross-validation RMSE. It had also been successfully integrated into the Streamlit application as a complete preprocessing and prediction pipeline.

The Random Forest model was therefore retained because replacing it with XGBoost would have required additional implementation work without producing a meaningful improvement in predictive performance.

The deployed model explained only approximately 0.6% of the variation in unseen crop yields. It was therefore deployed to demonstrate the technical workflow rather than to provide reliable agronomic recommendations.

7.2 Streamlit Application

Streamlit was selected because it provides a straightforward method for converting a Python machine-learning workflow into an interactive web application.

The application receives the seven original predictor variables:

- crop type;
- rainfall;
- temperature;
- fertilizer;
- nitrogen;
- phosphorus; and
- potassium.

Users are not required to calculate or enter the engineered variables manually.

After the original values are submitted, the application automatically creates:

- total NPK;
- nitrogen proportion;
- phosphorus proportion;
- temperature squared; and
- rainfall–fertilizer interaction.

The seven original predictors and five engineered predictors are then passed to the fitted Random Forest pipeline.

7.3 Application Workflow

The application follows the workflow shown below:

User input ⟶ Input validation ⟶ Feature engineering ⟶ Categorical encoding ⟶ Random Forest

The main stages are:

1. The user selects a crop type.
2. The user enters rainfall, temperature, fertilizer, nitrogen, phosphorus and potassium values.
3. The application validates that the values fall within the accepted numerical ranges.
4. The reusable feature-engineering function creates the five engineered predictors.
5. The fitted preprocessing pipeline one-hot encodes crop type.
6. The numerical and encoded categorical variables are passed to the Random Forest model.
7. The model returns the estimated crop yield in quintals per acre.

7.4 Reusable Feature-Engineering Module

The feature-engineering logic was stored in:

`src/feature_engineering.py`

This module contains the same formulas used during data preparation, model training and deployment.

Using a reusable module reduced the risk of calculating the engineered variables differently across the modelling and deployment stages.

The same feature-engineering function was used to:

- create the engineered dataset;
- prepare model inputs during testing;
- verify the exported model; and
- transform user inputs inside the Streamlit application.

This ensured consistency between the training and deployment environments.

7.5 Model Export

The complete fitted Random Forest pipeline was exported using the `joblib` library.

The complete pipeline was saved rather than only the Random Forest estimator. This preserved the categorical encoding and ensured that the same preprocessing steps would be applied during deployment.

The saved bundle contained:

- the fitted preprocessing pipeline;
- the trained Random Forest model;
- the original feature names;
- the engineered feature names;
- the expected model-input order;
- crop-type categories;
- numerical input ranges;
- target-variable information;
- model-performance results; and
- the installed scikit-learn version.

Two model files were saved:

```
models/rf_engineered_bundle.joblib
models/deployment_model.joblib
```

The first file preserves the selected Random Forest model under a descriptive filename. The second represents the model currently loaded by the Streamlit application.

This structure allows the deployment model to be replaced in the future without redesigning the application interface.

7.6 Saved-Model Verification

The saved model bundle was reloaded before deployment.

A sample observation containing the seven original predictors was passed through the reusable feature-engineering function to create the complete model input.

The prediction from the reloaded model was compared with the prediction produced by the fitted model still available in the notebook.

The verification test passed successfully, confirming that:

- the model bundle had been saved correctly;
- the model could be reloaded successfully;
- the expected feature order was preserved;
- the preprocessing pipeline remained available; and

- the reloaded model reproduced the notebook prediction.

For the sample observation used during verification, the estimated crop yield was:

10.054 quintals per acre.

7.7 Dashboard Features

The Streamlit dashboard contains:

- a crop-type selection field;
- numerical input fields for rainfall, temperature, fertilizer and nutrient values;
- a button for generating a prediction;
- the estimated crop yield;
- an expandable section showing the generated engineered features;
- the name of the current deployment model;
- the model's test MAE, RMSE and R^2; and
- a warning describing the model's limited predictive performance.

The numerical input fields use ranges obtained from the complete processed modelling dataset. This helps prevent users from entering values that are far outside the ranges observed during model development.

However, remaining within the observed ranges does not guarantee that a prediction will be accurate or appropriate for a specific farm, location or growing season.

7.8 User Interface

Figure 12 shows the application interface for entering the required crop and environmental information.

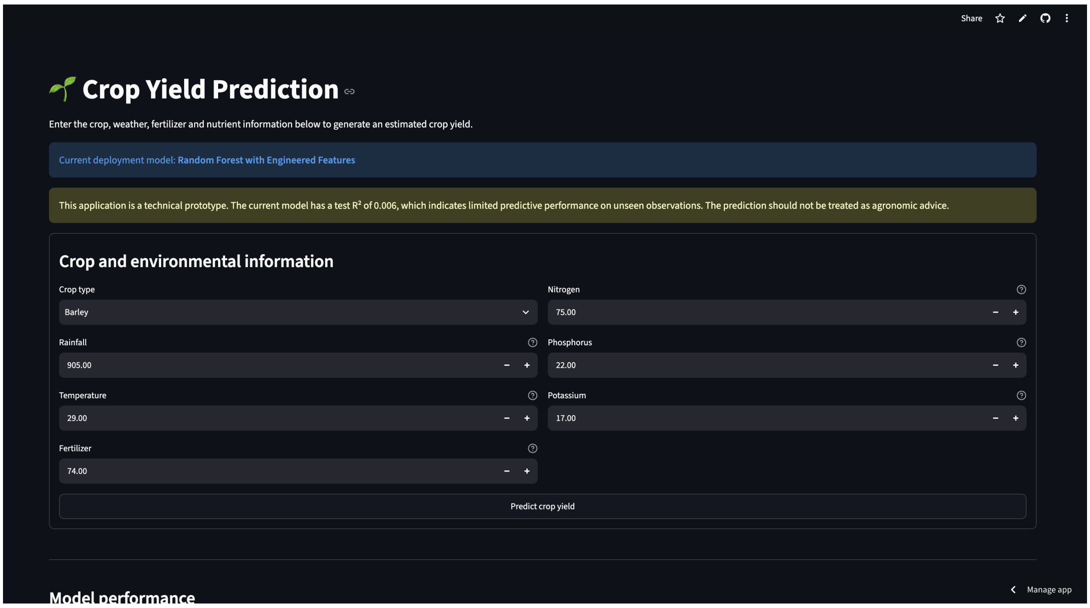

Figure 12: Streamlit interface for entering crop and environmental information

Figure 13 shows an example prediction generated by the application.

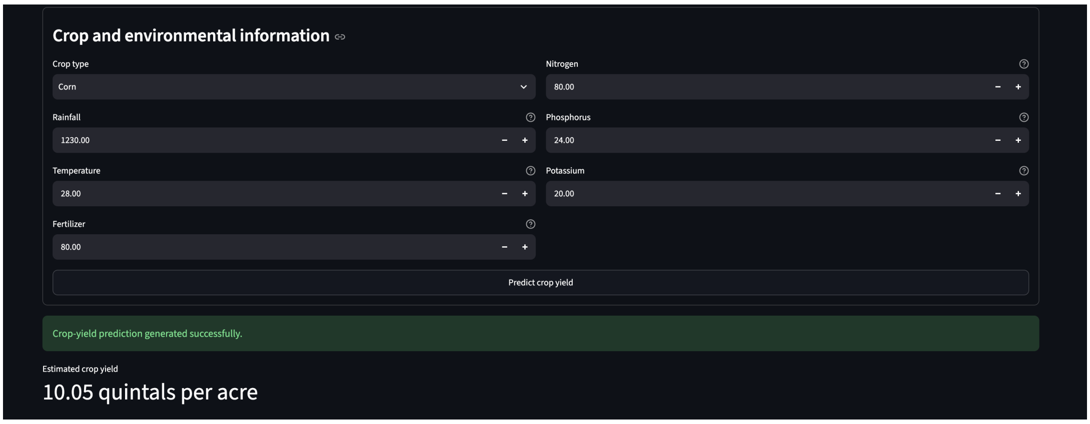

Figure 13: Example crop-yield prediction generated by the Streamlit application

These screenshots demonstrate the main user workflow without reproducing every application screen.

7.9 Local Testing

The application was first tested locally using:

```
streamlit run streamlit_app.py
```

The local test confirmed that:

- the exported model loaded successfully;
- all input fields functioned correctly;
- the engineered variables were created automatically;

- the prediction button generated an output;
- the displayed model metrics were correct; and
- the Streamlit prediction matched the notebook prediction for the same input values.

The successful local test confirmed that the model, feature-engineering module and interface were integrated correctly.

7.10 Repository and Deployment Files

The main deployment files were organised as follows:

```
streamlit_app.py
requirements.txt
models/deployment_model.joblib
models/rf_engineered_bundle.joblib
src/__init__.py
src/feature_engineering.py
```

The `requirements.txt` file contains the Python packages required to reproduce the application environment.

The project files were stored in the team GitHub repository:

`https://github.com/Ronkecrown/gamma2-crop-yield-prediction`

7.11 Model Disclaimer

A clear warning is displayed in the application because the selected model has limited predictive performance.

The dashboard informs users that:

- the application is a technical prototype;
- the deployed model has a test R^2 of 0.006;
- the dataset does not contain several important agricultural variables;
- the estimated yield should not be treated as agronomic advice; and
- the output should not be used as the sole basis for farming, lending, investment or resource-allocation decisions.

This disclaimer ensures that the model is presented honestly and consistently with the findings of the analysis.

7.12 Deployment Limitations

The successful implementation of the application demonstrates that the technical pipeline works from user input to prediction. It does not establish that the model is suitable for operational agricultural use.

The principal deployment limitations are:

- the deployed model has weak explanatory and predictive performance;
- the training data lack location, season, soil and farm-management information;
- the model has not been externally validated using independent agricultural data;
- the prediction does not include an uncertainty interval; and
- the application has not been evaluated with farmers, agricultural extension workers or other intended users.

The application should therefore be regarded as a proof of concept for the deployment process.

7.13 Potential Future Deployment Improvements

The application can be improved in future by:

- replacing the current model after more informative data and better-performing models become available;
- adding crop-specific prediction models;
- including location, season, soil and irrigation inputs;
- providing prediction intervals or uncertainty estimates;
- recording user inputs and verified outcomes for future model improvement;
- improving the visual design and accessibility of the interface;
- conducting usability testing with intended users; and
- validating the system using independent agricultural data.

Because the application loads a generic deployment model file, a future model may replace the current Random Forest without requiring major changes to the user interface.

8 Limitations and Future Work

The findings of this project should be interpreted within the limitations of the available dataset, modelling process, model-interpretation methods and deployment environment.

8.1 Dataset Limitations

The main limitation was the restricted predictive information contained in the supplied dataset.

The dataset included rainfall, temperature, fertilizer, nitrogen, phosphorus, potassium, crop type and yield. However, crop yield is influenced by several additional environmental, biological and management factors that were not recorded.

Important missing variables included:

- geographic location;
- production year and farming season;
- planting and harvesting dates;
- soil type;
- soil pH;
- soil moisture;
- organic-matter content;
- irrigation practices;
- crop variety;
- fertilizer type and application timing;
- pest and disease conditions; and
- farm-management practices.

The absence of these variables may explain why the available predictors accounted for only a very small proportion of the observed variation in crop yield.

8.1.1 Sample Size

After removing one duplicate observation, the cleaned dataset contained 999 records.

Although this sample size was sufficient for exploratory analysis and the development of prototype models, it may be limited for identifying complex nonlinear relationships, crop-specific effects and interactions among environmental and management variables.

A larger dataset covering different farms, regions, crop varieties, seasons and production years would provide more information for training and evaluating machine-learning models.

8.1.2 Lack of Geographic and Temporal Identifiers

The dataset did not contain location, date, season or production-year information.

This prevented the analysis from accounting for spatial and temporal differences in crop production. It also prevented reliable integration of external weather, soil or satellite data.

Adding external information without reliable geographic and temporal matching identifiers could have assigned incorrect environmental conditions to the observations and introduced additional error into the analysis.

8.1.3 Measurement Information

The project brief provided units for rainfall, temperature, fertilizer and crop yield.

However, the exact measurement units, collection procedures and recording conditions for nitrogen, phosphorus and potassium were not fully documented.

Differences in measurement procedures or data-collection conditions may affect the interpretation and comparability of these variables.

8.2 Exploratory Analysis Limitations

The exploratory analysis examined associations between individual predictors and crop yield.

These analyses were useful for identifying patterns, distributions and possible relationships, but they do not establish causal effects.

For example, a positive association between rainfall and crop yield would not prove that increasing rainfall would directly cause yield to increase. The observed relationship could also be influenced by location, soil characteristics, crop variety, season or farm-management practices that were not included in the dataset.

Most predictor relationships were weak, which limited the conclusions that could be drawn from the exploratory analysis.

8.3 Feature-Engineering Limitations

The engineered variables were created from the original predictors. They included total NPK, nutrient proportions, temperature squared and the rainfall–fertilizer interaction.

Although these features represented reasonable agricultural and statistical relationships, they could not introduce information that was absent from the original dataset.

The engineered variables were also mathematically related to the original predictors. For example:

- total NPK was calculated from nitrogen, phosphorus and potassium;
- the nutrient proportions were calculated from the same nutrient variables;
- temperature squared was derived directly from temperature; and
- the rainfall–fertilizer interaction was calculated from rainfall and fertilizer.

These relationships introduced redundancy among some predictors. They may also divide feature importance among related variables and make individual feature effects more difficult to interpret.

The engineered variables produced only minor changes in predictive performance. This confirms that mathematical transformations alone could not compensate for the absence of important agricultural information.

8.4 Modelling Limitations

Three model families were evaluated:

- Multiple Linear Regression;

- Random Forest Regression; and
- Extreme Gradient Boosting.

Multiple Linear Regression may be limited when relationships are nonlinear or when predictors interact in complex ways.

Random Forest can capture nonlinear relationships and interactions, but it may learn patterns that do not generalise when the available variables contain weak predictive signals.

This was reflected in the performance of the engineered Random Forest. It achieved a training R^2 of 0.197 but a test R^2 of only 0.006. The difference indicates that some patterns learned from the training data did not generalise effectively to unseen observations.

XGBoost was also able to represent nonlinear relationships and interactions. However, it achieved a training R^2 of 0.0957 and a test R^2 of only 0.0091. Although it recorded a marginally lower hold-out test RMSE than the engineered Random Forest, the improvement was too small to be practically meaningful.

The weak results across linear regression, bagging and boosting approaches suggest that the main limitation was the information content of the dataset rather than the use of a particular modelling algorithm.

8.4.1 Limited Model Comparison

The completed analysis included three modelling algorithms and five modelling experiments:

1. Multiple Linear Regression using original predictors;
2. Multiple Linear Regression using engineered predictors;
3. Random Forest using original predictors;
4. Random Forest using engineered predictors; and
5. XGBoost using engineered predictors.

This comparison covered linear regression, bagging and boosting approaches, but it was not an exhaustive assessment of all possible regression methods.

Other algorithms may produce slightly different numerical results. However, the similar performance of the evaluated models and the consistently low test R^2 values suggest that changing the algorithm alone is unlikely to produce a major improvement without more informative data.

8.4.2 Sensitivity to the Train-Test Split

The hold-out test results depended partly on the observations assigned to the test set.

This was demonstrated when the raw and cleaned datasets were split separately using the same random state. Only 84 of the 200 test observations were common to both test sets.

Removing one observation changed the ordering and allocation of the remaining records during the split. Consequently, differences in test-set composition could affect MAE, RMSE and R^2.

The cleaned dataset and a consistent split convention were therefore used for the final model comparison. Five-fold cross-validation was also performed to provide a more stable assessment of model performance across different data subsets.

8.4.3 Hyperparameter-Tuning Limitations

Grid search was used to tune the Random Forest and XGBoost models. However, only a predefined and relatively small set of parameter combinations was evaluated.

The selected parameters were therefore the best values within the specified search spaces, not necessarily the best possible configurations.

A wider or more adaptive search method, such as randomised search or Bayesian optimisation, might identify different parameter values. Nevertheless, extensive tuning would be unlikely to overcome the limited predictive information contained in the current dataset.

8.4.4 Absence of an Independent Validation Dataset

The models were evaluated using a hold-out test set and cross-validation derived from the same source dataset.

No independent dataset from another farm, location, season or production period was available.

Therefore, the ability of the selected model to generalise to new farms, regions, crop varieties or growing seasons could not be confirmed.

External validation would be required before the model could be considered for operational use.

8.5 Feature-Impact Analysis Limitations

Impurity-based importance, permutation importance, partial dependence plots and crop-type scenario analysis were used to explain the engineered Random Forest.

The full impact analysis was limited to the Random Forest because it was retained as the deployment model. A separate impact analysis was not conducted for XGBoost because it was evaluated for predictive comparison but was not selected for deployment.

The interpretation methods describe the behaviour of the fitted Random Forest model. They do not represent confirmed agricultural or causal effects.

Impurity-based importance may favour continuous variables or variables with more possible split points.

Permutation importance may underestimate the contribution of correlated or mathematically related predictors because their importance can be shared among them.

Partial dependence plots show average model behaviour and may hide differences between individual observations, farms or crop categories. They may also represent unrealistic predictor combinations when variables are strongly related.

The crop-type scenario analysis changed crop type while keeping all other variables unchanged. Such combinations may not reflect realistic farming conditions for every crop.

Because the selected Random Forest achieved a test R^2 of only 0.006, all feature-impact findings should be interpreted cautiously. The results explain how the fitted model generated predictions, but they do not provide strong evidence of real-world agronomic effects.

8.6 Model-Selection Limitations

The engineered Random Forest was retained for deployment even though XGBoost recorded a marginally lower hold-out test RMSE and slightly higher test R^2.

The decision considered several evaluation criteria rather than relying on a single metric. Random Forest achieved the lower test MAE and a slightly lower mean cross-validation RMSE. It had also been successfully integrated into the Streamlit application.

The performance differences between the two models were negligible. Therefore, selecting Random Forest should not be interpreted as evidence that it was substantially superior to XGBoost.

Both models demonstrated weak predictive performance, and neither should be regarded as a reliable production model using the current dataset.

8.7 Deployment Limitations

The Streamlit application successfully demonstrated the complete workflow from user input and feature engineering to model prediction.

However, its limitations include:

- limited predictive performance of the deployed model;
- dependence on predictor ranges observed in the processed modelling dataset;
- absence of uncertainty estimates around individual predictions;
- lack of validation using independent agricultural data;
- no connection to live weather, soil or farm-management data;
- no evaluation across different geographic regions or production seasons; and
- no formal usability or agronomic validation with intended users and agricultural experts.

Restricting inputs to the ranges observed in the dataset reduces extreme extrapolation but does not guarantee that a prediction is accurate or appropriate for a particular farm.

The application should therefore be treated as a technical prototype and proof of concept.

It should not be used as the sole basis for farming, fertilizer, lending, financing, investment or resource-allocation decisions.

8.8 Future Work

Future work should focus first on improving the quality, coverage and relevance of the dataset.

Recommended future steps include:

1. collect more observations across different farms, locations, seasons, production years and farming conditions;
2. include soil, irrigation, crop-variety, pest, disease and farm-management variables;

3. record geographic coordinates and reliable temporal identifiers;
4. establish clear measurement units and standardised data-collection procedures;
5. integrate external weather, soil and satellite data only when reliable matching identifiers are available;
6. include a dummy mean regressor as a formal benchmark for future modelling experiments;
7. evaluate models separately by crop type when sufficient observations are available for each category;
8. compare additional regression algorithms after the dataset has been improved;
9. consider crop-specific, spatial or spatiotemporal modelling approaches when suitable identifiers become available;
10. use repeated cross-validation or nested cross-validation to assess model stability and reduce optimistic model-selection bias;
11. provide prediction intervals or other measures of predictive uncertainty;
12. validate the selected model using independent agricultural data from different locations and production periods;
13. conduct usability testing with intended dashboard users; and
14. involve agricultural experts in reviewing the variables, feature engineering, model outputs and practical recommendations.

The main future priority should be improving data quality and relevance rather than relying only on increasingly complex algorithms. More sophisticated models will be useful only when the dataset contains sufficient information to represent the major factors that influence crop yield.

9 Conclusion

This project developed and evaluated machine-learning models for predicting crop yield using rainfall, temperature, fertilizer application, nitrogen, phosphorus, potassium and crop type.

The project addressed three main tasks:

1. predicting crop yield using the supplied dataset;
2. analysing the contribution of the available variables to model predictions; and
3. providing recommendations for the firm based on the findings.

The original dataset contained 1,000 observations and eight variables. One duplicate observation was removed during data cleaning, resulting in a final cleaned dataset of 999 observations.

Exploratory data analysis showed that crop yield was approximately symmetrically distributed and contained no apparent extreme outliers. The crop categories were also reasonably balanced.

However, the available predictors showed generally weak relationships with crop yield. Rainfall had a statistically significant but weak positive association with yield, while temperature showed a weak and mildly nonlinear relationship. Fertilizer, nitrogen, phosphorus and potassium showed weak or negligible standalone relationships with yield. Crop type also explained very little variation in yield.

Five engineered variables were created to investigate whether additional patterns could be captured:

- total NPK;
- nitrogen proportion;
- phosphorus proportion;
- temperature squared; and
- rainfall–fertilizer interaction.

Three model families were evaluated:

- Multiple Linear Regression;
- Random Forest Regression; and
- Extreme Gradient Boosting.

Multiple Linear Regression and Random Forest were each evaluated using the original and engineered predictor sets. XGBoost was evaluated using the engineered predictor set. This produced five modelling experiments in total.

The Multiple Linear Regression models produced test R^2 values close to zero. This indicated that the available predictors explained very little variation in unseen crop yields under a linear modelling assumption.

The Random Forest models captured more variation in the training data but did not generalise strongly to the test data. The engineered Random Forest achieved:

- test MAE of 2.525 quintals per acre;
- test RMSE of 2.926 quintals per acre;
- test R^2 of 0.006; and
- five-fold cross-validation RMSE of 2.762 ± 0.125.

The engineered XGBoost model achieved:

- test MAE of 2.5395 quintals per acre;

- test RMSE of 2.9219 quintals per acre;
- test R^2 of 0.0091; and
- five-fold cross-validation RMSE of 2.7699 ± 0.1183.

XGBoost recorded the lowest hold-out test RMSE and the highest test R^2. However, its reduction in test RMSE compared with the engineered Random Forest was only 0.0041 quintals per acre. This difference was too small to represent a meaningful practical improvement.

The engineered Random Forest achieved the lowest test MAE and a slightly lower mean cross-validation RMSE than XGBoost. It had also been successfully integrated into the Streamlit application as a complete preprocessing and prediction pipeline.

The engineered Random Forest was therefore retained as the deployment model. This decision was based on its overall validation performance, lower average absolute error and successful technical implementation rather than on a claim that it substantially outperformed XGBoost.

The highest test R^2 among all five experiments was only 0.0091. This means that even the strongest hold-out result explained less than 1% of the variation in unseen crop yields. None of the evaluated models therefore demonstrated strong predictive performance.

The feature-impact analysis was conducted for the engineered Random Forest because it was retained as the deployment model.

Impurity-based importance, permutation importance, partial dependence plots and crop-type scenario analysis were used to examine the behaviour of the fitted model. The analysis suggested that rainfall and temperature were relatively more useful than the remaining predictors. However, their absolute predictive contributions were small.

These interpretation results describe how the fitted Random Forest generated predictions. They should not be interpreted as evidence that the identified variables have causal agricultural effects.

The limited model performance was consistent with the exploratory findings. The weak results across linear regression, bagging and boosting approaches suggest that the main challenge was not simply the choice of algorithm.

Instead, the supplied dataset lacked several important variables that may strongly influence crop yield. These included:

- geographic location;
- production year and season;
- soil characteristics;
- irrigation practices;
- crop variety;
- planting and harvesting dates;
- pest and disease conditions; and
- farm-management practices.

External data were not added because the dataset lacked the geographic and temporal identifiers required for reliable integration. Adding weather, soil or satellite information without accurate matching variables could have introduced unsupported assumptions and reduced the scientific validity of the analysis.

The engineered Random Forest was deployed through a Streamlit application. The application accepts the seven original predictors, automatically creates the five engineered variables and returns an estimated crop yield in quintals per acre.

The exported model was successfully reloaded and verified before deployment. The Streamlit application also ran successfully in the local environment, demonstrating that the model, feature-engineering module and user interface were technically integrated.

However, the application is presented as a technical prototype rather than a validated agricultural decision-support system. Its predictions should not be used as the sole basis for farming, fertilizer, lending, financing or investment decisions.

The main recommendation for the firm is to prioritise the collection of richer, more representative and better-documented agricultural data. Future datasets should include geographic, temporal, soil, irrigation, crop-variety, pest, disease and farm-management information.

Future model development should also include external validation, prediction intervals, crop-specific analysis and repeated or nested cross-validation after more informative data have been collected.

A reliable crop-yield prediction system will depend more on the quality, coverage and relevance of the available data than on the use of increasingly complex algorithms.

Overall, the project completed the full data-science workflow, including data preparation, exploratory analysis, feature engineering, model development, evaluation, interpretation and deployment.

The most important finding was that the available variables contained insufficient information for reliable crop-yield prediction. Reporting this result honestly provides a stronger scientific conclusion than presenting an artificially improved model without adequate evidence or justification.